

Abstract: T Type Multilevel Inverter

Multilevel inverters (MLIs) are essential power electronic converters that convert direct current (DC) from renewable energy sources, such as solar photovoltaic (PV) panels and fuel cells, into high-quality alternating current (AC) output. They are favored in medium- and high-power applications because they produce nearly sinusoidal voltage waveforms, significantly reducing total harmonic distortion (THD), minimizing electromagnetic interference (EMI), and improving voltage handling while lowering switching stress. Several MLI topologies have been developed over the years, including Neutral Point Clamped (NPC), Flying Capacitor (FC), Diode Clamped, and Cascaded H-Bridge (CHB). However, many conventional topologies encounter challenges such as increased complexity, larger sizes, more switches, and the need for multiple isolated DC sources, making them less ideal for compact or budget-sensitive applications.

The project focuses on single-phase T-Type multilevel inverters (T-MLIs), known for their structural simplicity, modular design, and fewer switching components compared to NPC or CHB topologies. T-MLIs can generate multiple voltage levels from a single DC source with fewer switches, leading to improved efficiency and easier control. The study explores both basic and advanced T-MLI configurations and assesses their performance using various modulation strategies. We specifically analyze modulation techniques such as Sinusoidal Pulse Width Modulation (SPWM) and Selective Harmonic Elimination (SHE). These techniques are crucial for shaping the output waveform and minimizing harmonic content. To enhance the efficiency and precision of harmonic elimination, we propose an optimization-based control method. The SHE-PWM technique is integrated with Particle Swarm Optimization (PSO), an evolutionary algorithm recognized for its rapid convergence and adaptability in real-time systems.

Additionally, we present a novel design for a 5-level flying capacitor multilevel inverter (FC-MLI), which addresses common issues encountered in traditional FC topologies. This proposed FC inverter minimizes the number of switches, capacitors, and associated gate drivers while ensuring voltage balance and maintaining the quality of the output waveform. Simulation results indicate that this streamlined topology not only enhances power conversion efficiency but also reduces overall costs and hardware complexity, making it especially suitable for compact renewable energy converters and standalone AC systems. This work contributes to the development of cost-effective, reliable, and scalable multilevel inverter systems, emphasizing low THD, diminished switching losses, and simplified control schemes. Such innovations will support the future of smart grids, microgrids, and off-grid renewable energy applications.

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